

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
3rd Semester of B. Pharm. Examination
University Theory Examination March 2018
PH231/PH219 Pharmaceutical Engineering-III

Date: 27/03/2018, Tuesday Time: 01:30 p.m. to 04:30 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION – I

Q 1 Attempt all questions. Each question is of one mark. 20

1. Size reduction is also known as
 - A. Maceration
 - B. Communion
 - C. Hammer mill
 - D. Fitz mill
2. _____ substances are mostly affected during milling due to heat generated
 - A. Thermolabile
 - B. Thermostable
 - C. Fine powdered
 - D. Fiber
3. Particle size of suspension is determined by
 - A. Sieving
 - B. Andersen pipette
 - C. Optical microscopy
 - D. Scanning electron microscope
4. Sieve shaker is used for the purpose of
 - A. Size reduction
 - B. Milling
 - C. Size separation
 - D. Mixing

5. _____ is a critical factor in formulating potent drugs
- A. Mixing
 - B. Temperature
 - C. Pressure
 - D. Polymorphism
6. Degree of mixing can also be referred as degree of _____
- A. Amperocity
 - B. Heterogeneity
 - C. Homogeneity
 - D. Segregation
7. Colloid mill is used for the preparation of
- A. Syrup
 - B. Elixir
 - C. Mouthwash
 - D. Emulsion
8. _____ causes entrapment of air
- A. Vortex
 - B. Diffusion
 - C. Current
 - D. Vanes
9. _____ is designed to improve the accuracy and efficiency of the Pharma industry process.
- A. Machinery
 - B. Automation
 - C. C.P.U.
 - D. Controller
10. Computer process control does not measure _____
- A. Operator error
 - B. Temperature
 - C. Pressure
 - D. Flow rate
11. Crystallization process is used for _____
- A. Extraction
 - B. Evaporation
 - C. Purification
 - D. Identification
12. Krystal Cooling Crystallizer is also known as _____ crystallizer
- A. Vacuum
 - B. Swenson Walker
 - C. Oslo
 - D. Agitated tank

13. Substance having the same crystalline form is called _____
- A. Isomorphs
 - B. Isobars
 - C. Isotopes
 - D. Isocrystals
14. In cyclone separator, the separation process depends on _____
- A. Size and density
 - B. Porosity
 - C. Melting point
 - D. Crystal type
15. The ratio of initial particle size to final particle size is defined as _____
- A. Reduction ratio
 - B. Kick's ratio
 - C. Rittingers ratio
 - D. Bond's ratio
16. Crystal structure can be studied by _____
- A. UV
 - B. HPLC
 - C. HPTLC
 - D. X-ray diffraction
17. Which one of the following is the majorly find route of exposure to hazardous drugs in pharmaceutical industry?
- A. Inhalation of an aerosolized drug
 - B. Dermal absorption
 - C. Ingestion
 - D. Injection
18. A process of crystallization taking place from vapor of a substance is known as _____
- A. Sublimation
 - B. Distillation
 - C. Precipitation
 - D. Filtration
19. Propellers, turbines and paddles are
- A. Vanes
 - B. Baffles
 - C. Impeller
 - D. Shaft
20. _____ is the mechanism of fluid energy mill.
- A. Inter-particulate attrition
 - B. Intra-particulate attrition
 - C. Glaring
 - D. Glazing

SECTION – II

Q 2 Attempt any TWO of the following.

- A Describe various factors affecting size reduction of pharmaceuticals. 05
- B Give principle, working and uses of Fluid energy mill with labelled diagram. 05
- C Describe various mechanism of size reduction with suitable example. 05

Q 3 Attempt any TWO of the following.

- A Explain theory for crystallization with statement of Mier's super saturation theory. 05
- B What do you mean by Caking of crystals? How it is prevented? 05
- C Give application of spherical crystallization in pharmacy. 05

SECTION – III

Q 4 Attempt any FOUR of the following.

- A Write a short note on cyclone separator. 05
- B Describe standards of sieve. Give working of Sieve shaker with diagram. 05
- C Describe principle, working and uses of Colloid mill with labelled diagram. 05
- D Write a note on pharmaceutical hazards. 05
- E Describe various safety precautions to be require in pharmacy laboratory. 05
- F Discuss various application of crystallization in pharmacy. 05

Q 5 Attempt any FOUR of the following.

- A How vortex formation is adversely affect the process of mixing? Explain. How it is prevented? 05
 - B Enlist mixers for semisolids mixing. Give principle, labeled diagram and advantages of planetary mixer. 05
 - C Explain the theory of solid-solid mixing. 05
 - D Give principle, working, labeled diagram and advantages of Impeller mixers. 05
 - E Classify industrial hazard. Briefly explain Carcinogens. 05
 - F Explain the elements of automatic process control systems. 05
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